Problems 2 | Output | Debug Console | Terminal | Ports | Spell Checker 2 | DevDb | Console Ninja | pwsh

```
PS C:\Users\Laptop Solutions\Downloads\NAVTTC-A2> py student_performance.py
============================================================
STUDENT PERFORMANCE AND RESULT MANAGEMENT SYSTEM
============================================================

A. VARIABLES AND DATA TYPES
Name: Ali Ahmad - <class 'str'>
Age: 20 - <class 'int'>
Enrolled: True - <class 'bool'>
Sample Mark: 85.5 - <class 'float'>

B. ENTER STUDENT INFORMATION
Enter student name: Adnan
Uppercase: ADNAN
Lowercase: adnan
Enter student age: 20
Enter roll number: 12
Enter admission year: 2026

C. PYTHON DATA STRUCTURES
Original subjects: ['English', 'Mathematics', 'Computer Science', 'Physics', 'Urdu']
After adding: ['English', 'Mathematics', 'Computer Science', 'Physics', 'Urdu', 'Biology']
After updating: ['English', 'Mathematics', 'Computer Science', 'Physics', 'Pakistan Studies', 'Biology']
After removing: ['English', 'Mathematics', 'Computer Science', 'Physics', 'Pakistan Studies']
Student tuple: ('12', 2026)
Unique subjects: ['English', 'Mathematics', 'Physics']
Student dictionary: {'name': 'Adnan', 'age': 20, 'marks': {}, 'roll_number': '12'}

D & E. ENTER SUBJECT MARKS

Subject 1: English
```

```
D & E. ENTER SUBJECT MARKS

Subject 1: English
Enter marks for English (0-100), or Q to stop: 80

Subject 2: Mathematics
Enter marks for Mathematics (0-100), or Q to stop: 84

Subject 3: Computer Science
Enter marks for Computer Science (0-100), or Q to stop: 90

Subject 4: Physics
Enter marks for Physics (0-100), or Q to stop: 92

Subject 5: Pakistan Studies
Enter marks for Pakistan Studies (0-100), or Q to stop: 83

==================================================
STUDENT PERFORMANCE REPORT
==================================================
Name: Adnan
Age: 20
Roll Number: 12
Admission Year: 2026

Subject-wise Marks
1. English: 80.00/100
2. Mathematics: 84.00/100
3. Computer Science: 90.00/100
4. Physics: 92.00/100
```

```
STUDENT AND SUBJECT TABLE
Student: Adnan
  English: 80.0
  Mathematics: 84.0
  Computer Science: 90.0
  Physics: 92.0
  Pakistan Studies: 83.0

Student record saved to student_records.txt

E. LOOP EXAMPLES
English
Mathematics
Computer Science
Physics
Pakistan Studies
1 English
2 Mathematics
3 Computer Science
4 Physics
5 Pakistan Studies
Valid sample mark: 45
Valid sample mark: 75
Valid sample mark: 90
Before stopping: 45
Before stopping: 75
Before stopping: -10

F. LIST COMPREHENSION AND ITERATORS
Passing marks: [78, 90, 62, 85]
Squares of even numbers: [4, 16, 36, 64, 100, 144, 196, 256, 324, 400]
```

```
============================================================
TESTING WITH THREE SETS OF MARKS
============================================================

Test Case 1
Marks: [90, 85, 88, 92, 80]
Total: 435/500
Percentage: 87.00%
Grade: A
Result: Pass

Test Case 2
Marks: [55, 60, 65, 70, 75]
Total: 325/500
Percentage: 65.00%
Grade: C
Result: Pass

Test Case 3
Marks: [95, 90, 40, 85, 90]
Total: 400/500
Percentage: 80.00%
Grade: Fail
Result: Fail

Program completed successfully.
PS C:\Users\Laptop Solutions\Downloads\NAVTTC-A2>
```